Fluorescence Resonance Energy Transfer (FRET)

황현석

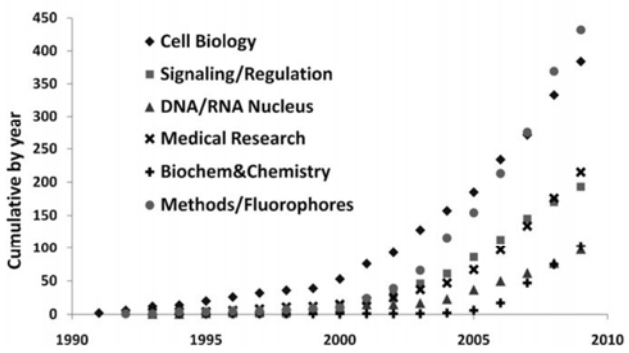
강원대학교 자연과학대학 화학과

Fluorescence resonance energy transfer

- Fluorescence resonance energy transfer (=Förster resonance energy transfer, FRET): 두 개의 형광 분자 사이에 짧은 거리 (10 nm 미만)에서 일어나는 에너지 이동 현상.

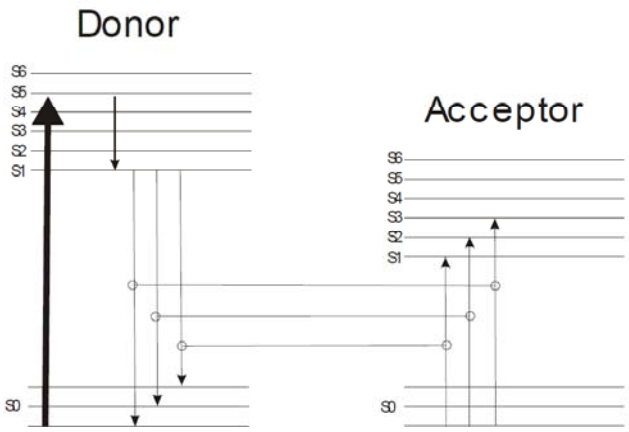


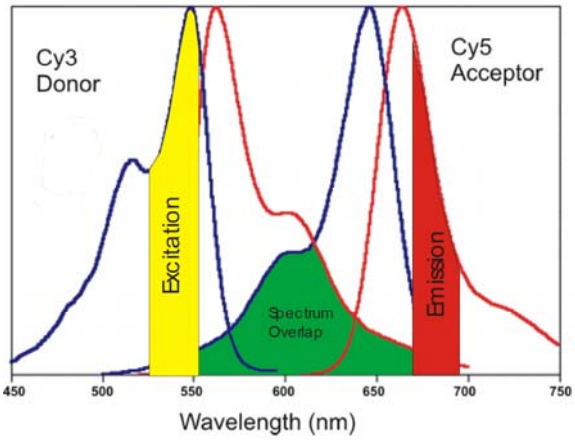
- 1946년에 Theodor Förster에 의해 FRET 현상이 발표 됨.
- Cumulative number of FRET publications by year in broad categories



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FRET mechanism



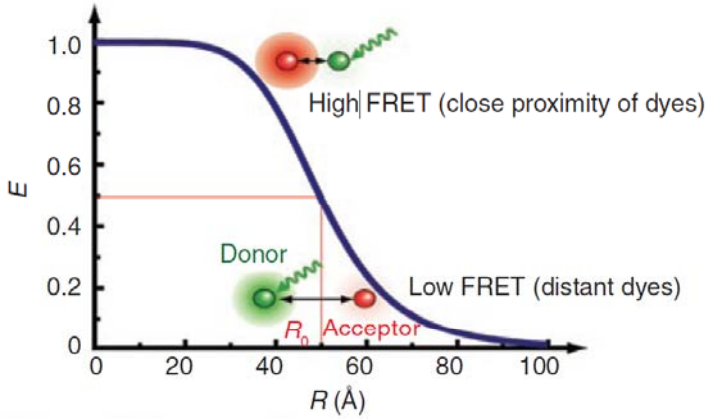


FRET efficiency

- FRET efficiency (E)

$$E = \frac{k_{FRET}}{k_{FRET} + k_R + k_{NR}} = \frac{1}{1 + (k_R + k_{NR})/k_{FRET}} = \frac{1}{1 + (R/R_0)^6}$$

  - k_{FRET}: FRET transfer rate
  - k_R: radiative decay rate
  - k_{NR}: nonradiative decay rate
  - R_0: Förster distance



Förster distance

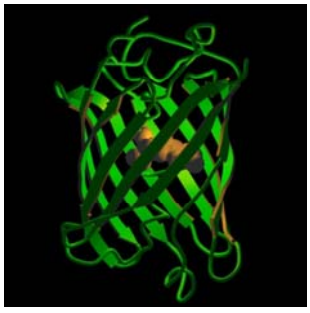
Table 1. Förster distances R_0 and photophysical properties for FP-based FRET pairs.

Donor–acceptor[a]	R_0[b] [nm]	FP photophysical properties[c] FP	QY	EC	Ex.	Em.
CFP–YFP	4.755	CFP	0.4	32.5	433	475
Cerulean–Venus	5.215	Cerulean	0.62	43	433	475
mTFP–Venus	5.701	mTFP	0.85	64	462	492
mTFP–mKO2	5.29	GFP	0.6	56	488	507
mTFP–tdTomato	6.201	YFP	0.61	83.4	514	527
mTFP–mTagRFP	5.058	Venus	0.57	92.2	515	528
mTFP–mKate	4.559	mOrange	0.69	71	548	562
GFP–mOrange	5.381	mKO2	0.62	63.8	551	565
GFP–mCherry	5.119	tdTomato	0.69	138	554	581
Venus–tdTomato	6.354	mTagRFP	0.41	81	558	584
Venus–mCherry	5.52	mCherry	0.22	72	587	610
Venus–mKate	5.042	mKate	0.33	45	589	636

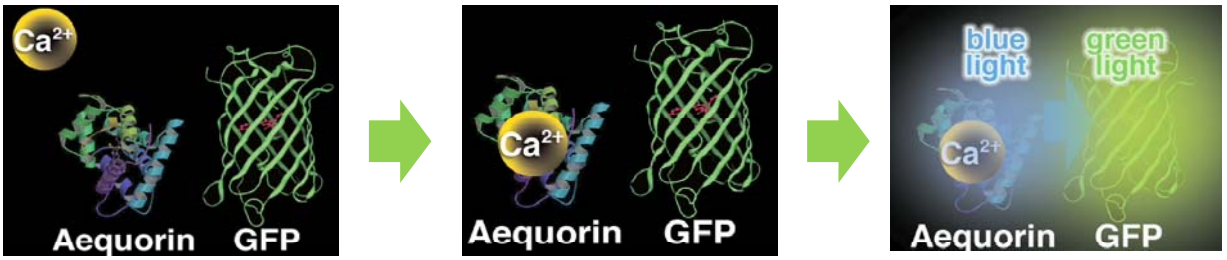
What is Green Fluorescent Protein (GFP)?



Aequorea victoria jellyfish

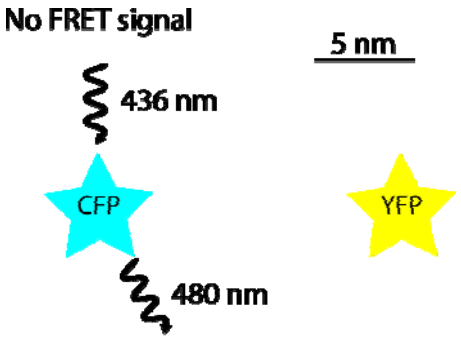


GFP

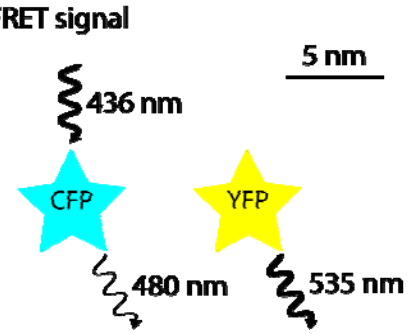


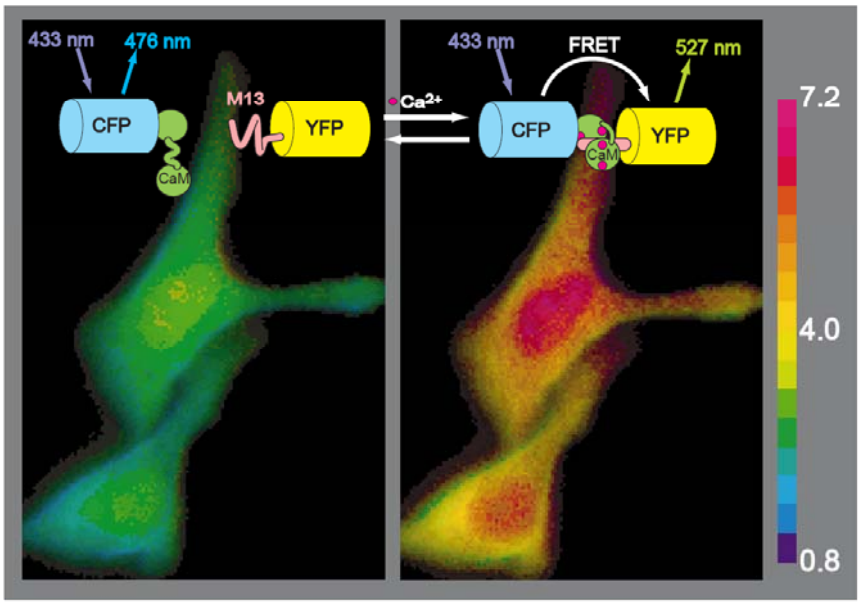
Cyan fluorescence protein (CFP) – yellow fluorescence protein (YFP) FRET

• CFP와 YFP의 거리가 5 nm 이상일 때



• CFP와 YFP의 거리가 5 nm 이하일 때

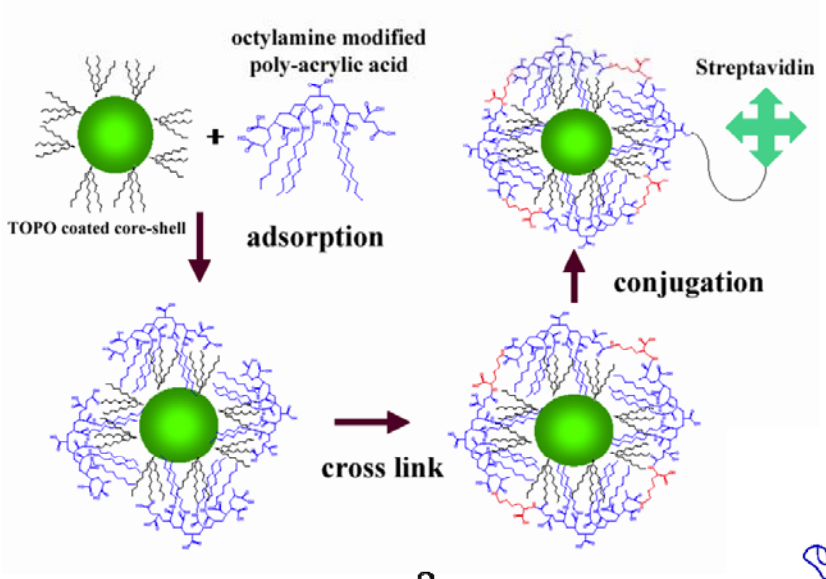


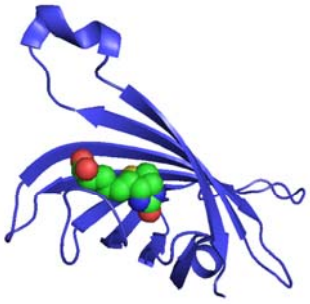


Cyan fluorescence protein (CFP) labeled calmodulin and yellow fluorescence protein (YFP) labeled calmodulin binding peptide (M13-YFP) were coexpressed. High Ca2+ levels (right) lead to binding and FRET emission of YFP (pseudo color red); low Ca2+ levels (left) lead to little FRET and mostly blue emission (pseudocolor green).

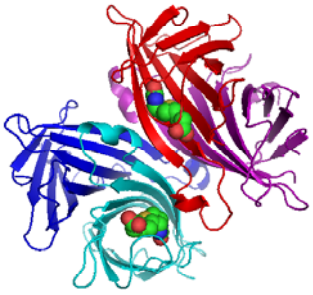
Tsien & Miyawaki, Science **280**, 1954–1955 (1998).

Biomolecule conjugation of quantum dots

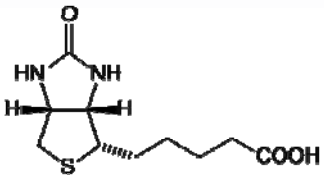




Steptavidin monomer
with a bound bzziotin



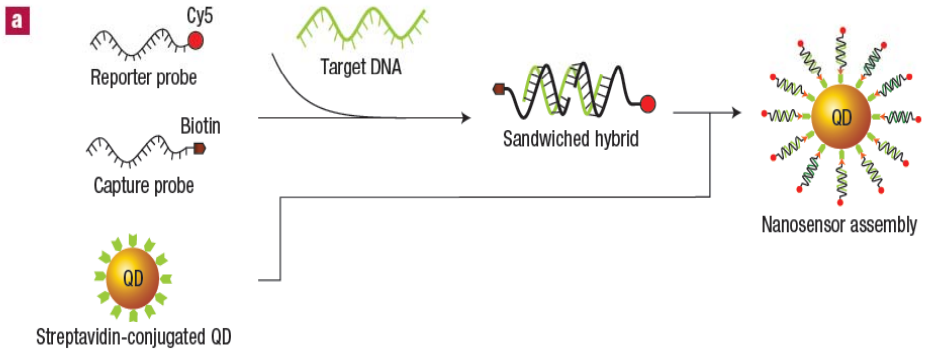
Steptavidin tetramer
with two bound biotins

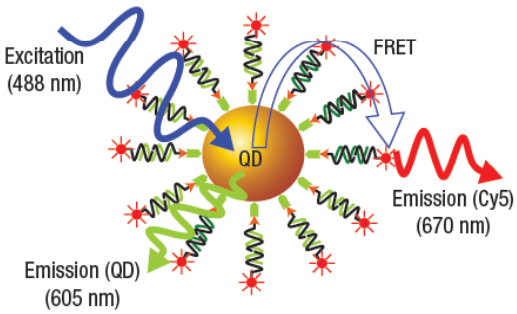


Biotin

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Single-quantum-dot-based DNA nanosensor





Zhang et al., Nature Mater. 4, 826 (2005)

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